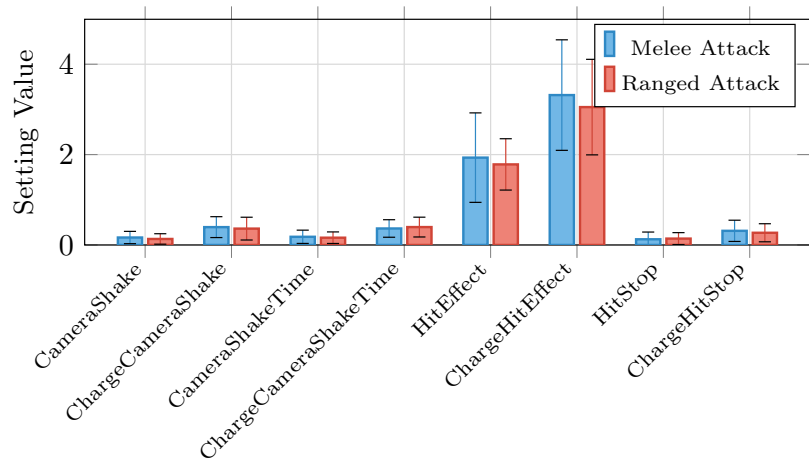


Quantifying Hit Feedback Preferences in 2D Action Games: A User-Adjustment Study of Visual and Temporal Parameters

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(C) Presentation Parameters

Figure 1: Experimental setup and findings. (A) Melee attack scene and (B) ranged attack scene where players adjust presentation parameters via sliders. (C) Comparison of presentation parameter settings shows no significant differences between attack distances (melee vs. ranged), indicating universal player preferences regardless of attack range.

Abstract

Hit feedback presentation at the moment of attack impact is fundamental to player experience in action games, yet quantitative understanding of optimal parameter settings remains limited. We present a user-adjustment methodology where players directly calibrate eight visual and temporal parameters in a 2D game. Through parameter adjustment ($N=16$) and gameplay validation ($N=116$) experiments, we quantified preferences across attack types and distances. Key findings reveal: (1) preferences are universal across attack distances but differ significantly by type, with charge attacks requiring $1.7\times$ stronger presentation; (2) individual differences span 6-fold ranges with three distinct preference types; (3) random settings are perceived as excessive 45% of the time, indicating comfortable ranges are biased toward lower values. Settings averaged from 16 participants achieved over 80% optimal ratings from

new users, demonstrating practical applicability. These findings provide evidence-based design guidelines emphasizing upper-bound constraints and customization options.

CCS Concepts

• **Human-centered computing** → **User centered design; Empirical studies in interaction design.**

Keywords

Hit feedback, Game feel, Player preferences, 2D action games, Player experience

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1 Introduction

Game development has become increasingly democratized, making it critical to provide engaging experiences within limited resources. Hit feedback presentation at the moment of attack impact is a



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fundamental element directly linked to players' sense of exhilaration and achievement. While the importance of sensory elements such as "control feel" and "tactility" is recognized, questions about what presentations work effectively and how to handle individual differences remain quantitatively unexplored.

In this study, we propose an experimental method where players adjust hit feedback presentations in 2D action games, and quantitatively analyze parameter setting tendencies and individual differences. In contrast to conventional research centered on subjective evaluation, we clarify the relationship between actual play behavior and presentation settings through analysis of behavioral data based on operation logs.

Specifically, we address the following research questions:

- **RQ1:** Are there differences in preferred presentation parameters depending on attack types (melee, ranged, normal, charge)?
- **RQ2:** What characteristics and patterns exist in players' presentation adjustment behaviors?
- **RQ3:** What are the important factors in presentation design considering individual differences?

Understanding how presentation elements interact with player psychology requires grounding in both game design theory and player motivation research. Swink's foundational work on "game feel" [16] established a vocabulary for describing the tactile and kinesthetic sensations in games, while research on player motivations [2, 18] has demonstrated how players differ in their motivations for achievement, social interaction, and immersion—differences that may extend to presentation preferences.

2 Related Work

Game presentation research has developed primarily from three perspectives.

First is understanding the impact of presentation elements on psychological experience. Hicks et al. empirically verified the impact of "juiciness" on player experience through both visual [5] and audio [6] elements. Kao et al. demonstrated through large-scale experiments how Juicy Feedback affects players' motivation and enjoyment through psychological factors of effectance, competence, and curiosity [9]. Pichlmair and Johansen systematized the concept of Game Feel into specific design elements [14].

Second is knowledge on presentation design in attack actions. Ujitoko and Ban comprehensively compiled the mechanisms of "pseudo-haptics," where visual operation delays such as cursor deceleration and hit-stop generate pseudo-tactile sensations [17]. These findings have been applied to VR environments [1, 11, 12]. Lécuyer's comprehensive survey [10] further established pseudo-haptics as a technique where visual feedback manipulates haptic perception, demonstrating that control-display ratio modifications can create illusory sensations without physical force feedback devices. As Interrante et al. showed, since distance perception in virtual space depends on visual cues [8], optimal feedback intensity is expected to differ by attack distance. Young et al. verified the relationship between time and reward in charge attacks [19].

Third are attempts at user-adjustable presentation. Bhatnagar et al. integrated "emotional impact" and "Game Feel" [3]. Garver et al. investigated the impact of visual style on player experience [4].

Table 1: Setting ranges and units of presentation parameters

Parameter	Range	Unit	Description
CameraShake	0.0–1.0	Unit	Camera shake intensity
CameraShakeTime	0.0–1.0	sec	Camera shake duration
HitEffect	0.0–5.0	Scale	Effect size (1.0 = standard)
HitStop	0.0–1.0	sec	Freeze time on hit

From a methodological perspective, formal frameworks for game design [7] and user-centered approaches [13] emphasize the importance of player feedback in iterative design. However, accessibility research [15] has highlighted that even well-intentioned design processes often fail to accommodate diverse player needs, particularly when relying solely on subjective evaluation rather than behavioral data. This underscores the value of methods that directly capture player adjustment behaviors.

However, much knowledge relies on subjective evaluations, and quantitative analysis based on operation logs and comparative verification by attack types are insufficient. This study quantitatively elucidates presentation setting tendencies and individual differences in 2D action games through slider adjustments by players themselves.

3 Method

We conducted two experiments in this study. In Experiment 1, we recorded the process of players freely adjusting presentation parameters to explore their preferred settings. In Experiment 2, we verified the effectiveness of the setting values obtained in Experiment 1 in an actual gameplay environment.

3.1 Experiment 1: Parameter Adjustment

We prepared a 2D action game environment built on Unity, with a melee attack scene (Fig. 1 (A)) and a ranged attack scene (Fig. 1 (B)). Each scene implements two types of attacks: normal and charge, and participants can adjust eight presentation parameters using sliders (Table 1). The experiment was conducted online, with participants running a distributed Unity application on their home PC environments. The frame rate was fixed at 60fps to maintain uniformity in physics calculations.

Participants performed the following four tasks in order: (1) Melee attack, pleasant condition: Adjust to settings that feel most suitable (pleasant); (2) Melee attack, unpleasant boundary condition: Adjust to settings that are just barely not unpleasant; (3) Ranged attack, pleasant condition: Adjust to optimal settings similar to melee; (4) Ranged attack, unpleasant boundary condition: Adjust to boundary settings similar to melee. When satisfied with the settings in each scene, participants pressed a button in the lower right to proceed, with no time limit set.

A total of 16 participants (14 male, 2 female, average age 22.5 years) including students from an information science university and external collaborators participated. Gaming frequency was "almost every day" for 11, "several times a week" for 3, and "several times a month" for 2, with the majority being regular gamers.

3.2 Experiment 2: Validation in Gameplay

To verify whether the setting values obtained in Experiment 1 function effectively in actual gameplay situations, we constructed a new stage-format 2D action game. Participants played three scenes (A, B, C) in randomized order and responded to an evaluation questionnaire about their play experience using Google Forms after each scene.

Presentation parameters for each scene were set as shown in Table 2. Scene A applied the mean values from the “pleasant condition” of Experiment 1. Scene B determined all parameters randomly. To create “unpleasant” settings for Scene C,

$$V_{\text{unpleasant}} = 2V_{\text{boundary}} - V_{\text{pleasant}} \quad (1)$$

we used this equation, adding the difference between the unpleasant boundary value V_{boundary} and pleasant value V_{pleasant} to the boundary value, intentionally setting excessive presentation.

A total of 116 students (93 male, 23 female, average age 21.1 years) from an information science university participated. Since some participated in both melee and ranged attack experiments, the total number of valid responses was 130 (60 melee, 70 ranged). Subjective evaluations used a 5-point Likert scale.

4 Results and Discussion

Normality was confirmed by Shapiro-Wilk test for all parameters (all $p > .05$), allowing the use of parametric tests.

4.1 RQ1: Differences by Attack Types

T-tests showed no significant differences in all presentation parameters between melee and ranged attacks (all $p > 0.10$, Fig. 1 (C)). In contrast, as shown in Fig. 2, significant differences were confirmed for all parameters between normal and charge attacks (all $p < 0.01$, Cohen’s $d = 1.01$ – 1.81). Notably, HitEffect increased approximately 1.7-fold from 1.858 for normal attacks to 3.184 for charge attacks (Cohen’s $d = 1.28$), demonstrating the importance of visual impact for charge attacks.

This indicates that players’ criteria for comfort are universal regardless of attack distance, while they have clearly different presentation expectations for attack types (normal/charge). Practically, individual adjustments for attack distance are unnecessary, but charge attacks require approximately 1.7-fold (HitEffect) more powerful presentation. This finding enables developers to create distinction in presentation intensity based on statistical evidence without relying on empirical rules. Additionally, the finding that adjustments by attack distance are unnecessary greatly contributes to reducing development costs.

4.2 RQ2: Adjustment Behavior Patterns

Time-series analysis revealed three adjustment behavior patterns: (1) exploratory type: fine-tuning after major changes; (2) gradual type: stepwise value increases; (3) intuitive type: few changes after early determination. The average adjustment time was approximately 120 seconds, with a median of approximately 84 seconds.

The three observed adjustment patterns provide important implications for presentation adjustment interface design. Adaptive UI is suggested, such as high-precision sliders for exploratory types and preset functions for intuitive types.

4.3 RQ3: Individual Differences and Design Factors

As shown in Fig. 3 (a), the largest individual differences were observed in HitEffect (range: 0.643–4.036, approximately 6-fold). Other parameters showed relatively small individual differences.

K-means cluster analysis revealed three preference groups (Fig. 3 (b)): (1) Impact-oriented type (high HitEffect and CameraShake settings); (2) Tempo-oriented type (extremely short HitStop and CameraShakeTime settings); (3) Balanced type (all parameters set to intermediate values). The cluster number $k=3$ was determined through examination of the Elbow method and Silhouette coefficient.

Correlation analysis revealed moderate positive correlations between CameraShakeTime and HitStop ($r = 0.60$), and between CameraShakeTime and HitEffect ($r = 0.59$). The correlation between CameraShake and HitEffect was weak ($r = 0.16$).

The largest individual differences (approximately 6-fold) in HitEffect demonstrate the importance of customization features according to individual preferences. Meanwhile, cluster analysis clearly classified three preference types, providing guidelines for preset design.

4.4 Validation in Gameplay (Experiment 2)

A total of 116 participants (93 male, 23 female, average age 21.1 years) participated in Experiment 2, with 130 valid responses (60 melee, 70 ranged).

As shown in Fig. 4 (a), Scene A applying mean values from Experiment 1 was ranked first by 81.7% for melee and 72.9% for ranged attacks. In overall satisfaction (Fig. 4 (b)), Scene A was significantly higher than other scenes (both $p < 0.01$). No significant differences were found between Scene B (random) and Scene C (unpleasant).

The mean values from a small number of participants in Experiment 1 ($N=16$) were evaluated as “optimal” by over 80% of new participants ($N=130$). This demonstrates that the simple method of averaging adjustment results from several representative individuals is practical as an alternative to large-scale testing.

The most important finding is that random settings (Scene B) were as unpopular as unpleasant settings (Scene C), with the cause mainly being “excessive presentation” (45%). Mathematically, “insufficient” and “excessive” should occur to similar degrees, but “insufficient” was only 7% in reality. This means that comfortable parameter ranges are biased toward the lower side of the settable range, providing design guidelines that strict upper limits are necessary in automatic generation of presentations.

4.5 Limitations

Limitations of this study include: (1) small sample size in Experiment 1 ($N=16$), (2) participants mainly limited to information science students, and (3) focus only on 2D action games. Future work requires verification with larger and more diverse participants, and extension research to 3D games and FPS.

Table 2: Parameter settings for each scene in Experiment 2. Scene B: Uniformly distributed random values within full parameter ranges.

Parameter	Scene A (Pleasant)	Scene B (Random)	Scene C (Unpleasant)
<i>Melee Attack</i>			
HitStop	0.126	0.0–1.0	0.572
CameraShakeTime	0.181	0.0–1.0	0.539
CameraShake	0.165	0.0–1.0	0.513
HitEffect	1.933	0.0–5.0	2.973
<i>Ranged Attack</i>			
HitStop	0.141	0.0–1.0	0.511
CameraShakeTime	0.161	0.0–1.0	0.549
CameraShake	0.133	0.0–1.0	0.397
HitEffect	1.783	0.0–5.0	3.401

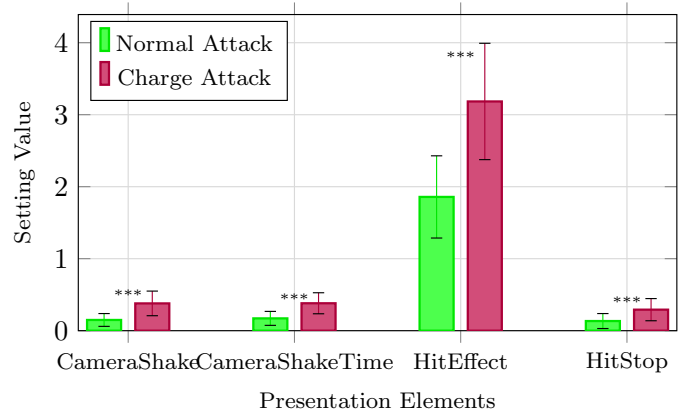


Figure 2: Comparison of presentation parameters between normal and charge attacks (mean $\pm$ SD). * $p < 0.01$.**

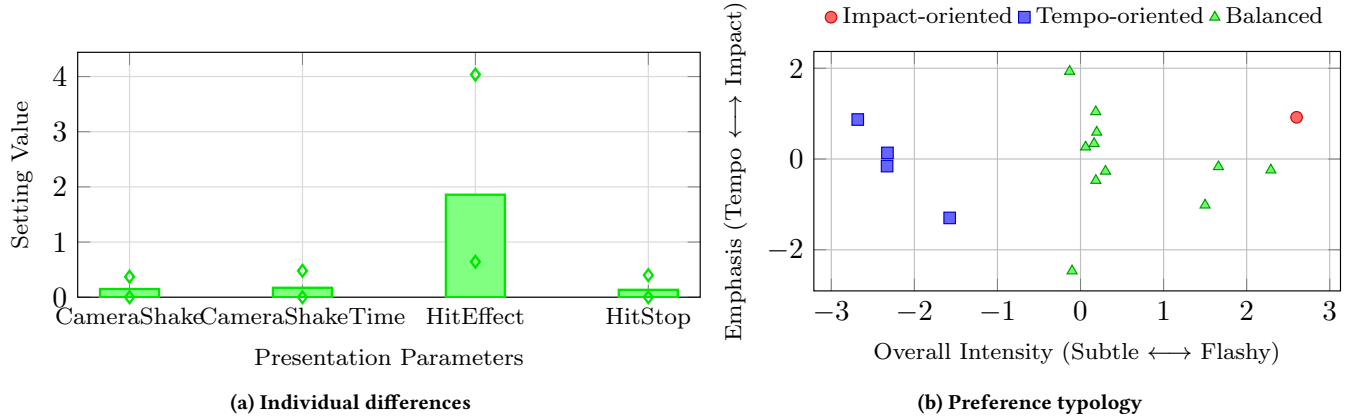


Figure 3: Individual differences and preference typology. (a) Distribution showing largest variance in HitEffect (6-fold range) with diamond markers indicating min/max values. (b) Three preference types identified through cluster analysis. Each point represents one participant (N=16). Axes derived from principal component analysis of four presentation parameters.

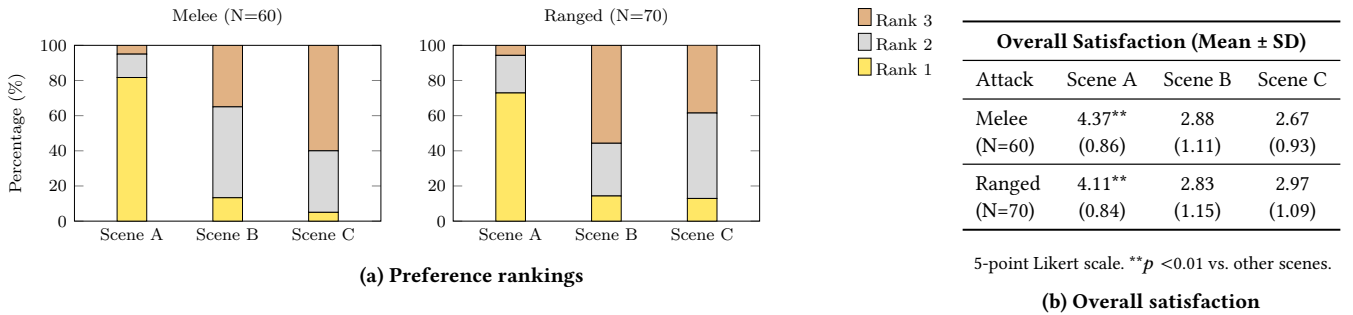


Figure 4: Validation experiment results. (a) Scene A (averaged settings from Experiment 1) was ranked first by 81.7% (melee) and 72.9% (ranged). (b) Scene A showed significantly higher satisfaction than random (Scene B) and unpleasant (Scene C) settings.

5 Conclusion

This study quantitatively elucidated design guidelines for attack hit feedback presentation in 2D action games through user-adjustment experiments.

Key findings for the three research questions are as follows: **RQ1:** No significant differences by attack distance (melee/ranged), but significant differences for all parameters by attack type (normal/charge) with large effect sizes (Cohen's $d = 1.01\text{--}1.81$). Charge attacks require approximately 1.7-fold setting values for visual effects. **RQ2:** Adjustment behaviors are classified into three patterns (exploratory, gradual, intuitive), providing guidelines for adaptive UI design. **RQ3:** Six-fold individual differences exist in HitEffect, with preferences classified into three types (Impact-oriented, Tempo-oriented, Balanced).

The most important finding is that random settings are perceived as excessive presentation (45%), indicating that comfortable ranges are biased toward the lower limit of the settable range. This demonstrates that strict upper limits are essential in automatic generation of presentations. Additionally, mean values from a small group ($N=16$) were evaluated as optimal by over 80% of new users ($N=130$), demonstrating the practicality of simple averaging methods.

As practical contributions, this study provides developers with: (1) statistically-grounded presentation ratios for charge/normal attacks (approximately 1.7-fold); (2) cost-reduction insight that adjustments by attack distance are unnecessary; (3) design principles for upper limit restrictions in automatic generation; (4) preset design guidelines based on three preference types.

Future research directions include verification with larger and more diverse participants, extension to 3D games and FPS, multi-modal verification including auditory elements, and construction of personalized adaptive presentation systems using machine learning.

Acknowledgments

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